Nonnegative and positive factorizations

Complexity of globally optimal nonnegative rank-two approximation

Difficulty: extreme **Importance:** broadly interesting

Topic: low-rank approximation; computational complexity

Status: open; admitted after a bounded literature search found no resolution.

Problem statement

Given an arbitrary $X \in \mathbb{Q}_+^{m\times n}$ and $\tau \in \mathbb{Q}_{\geq 0}$, determine the complexity of deciding whether

$$\exists W \in \mathbb{R}_+^{m\times 2},\ H \in \mathbb{R}_+^{2\times n} : \quad \sum_{i=1}^{m}\sum_{j=1}^{n}\left(X_{ij} - \sum_{k=1}^{2} W_{ik}H_{kj}\right)^2 \leq \tau.$$

In particular, is there a deterministic algorithm polynomial in the total binary input length, or is this decision problem NP-hard under polynomial-time many-one reductions? This is a complexity-classification question, without an assumption that these two outcomes exhaust the possibilities. The factors may have real entries; only the data and threshold must be rational. Their inner dimension is at most two, with a zero factor column permitted.

This fixes an exact decision interpretation of the literature's global rank-two NMF optimization question. There is no promise that X itself has rank two. When a rank-two truncated SVD of X is nonnegative, a best nonnegative rank-two approximation is obtainable from it. Arbitrary input can fall outside this tractable special case, and alternating nonnegative least squares need not find a global optimum.

References

Gillis, *Nonnegative Matrix Factorization*, §6.1.3, p. 199, Theorem 6.6 and final paragraph. Lindy, Noferini, and Van Dooren, *On rank-2 Nonnegative Matrix Factorizations and their variants*, §1, with §3's suboptimal approximation and §4's ANLS initialization.

Status check

Searches for *rank two nonnegative matrix factorization approximation NP hard polynomial time 2025 2026* and *rank-2 NMF complexity 2026* found no resolution. The July 2025 primary paper explicitly identifies rank two as an unresolved complexity case; its contribution is an effective initial approximation and heuristic refinement. The March 2026 constrained nonnegative Gram-feasibility preprint concerns partially specified symmetric matrices with affine side constraints, which are absent from the problem above.